

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

code

```
def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array) - i - 1):
            if array[j] > array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp
```

```
data = [-2, 45, 0, 11, -9]
bubbleSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("roshan")
```

output

```
===== RESTART: C:/Users/101
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
roshan
|
```

b. Insertion sort

code

```
def insertionSort(array):
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key
```

```
data = [9, 5, 1, 4, 3]
insertionSort(data)
```

```
print('Sorted Array in Ascending Order:')
print(data)
print("roshan ")
```

output

```
----- RESTART: C:/Users/roshan/Python36/python.exe
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
roshan
```

c. Selection sort

code

```
def selectionSort(array, size):
    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):
            if array[i] < array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])
```

```
data = [-2, 45, 0, 11, -9]
```

```
size = len(data)
```

```
selectionSort(data, size)
```

```
print('Sorted Array in Descending Order:')
```

```
print(data)
```

```
print("Roshan")
```

output

```
===== RESTART: C:/Users/itlab
```

```
Sorted Array in Ascending Order:
```

```
[1, 3, 4, 5, 9]
```

```
Roshan
```

```
|
```